

EXP.NO: 4	Create a web form that accepts a user's name and age. Write a Servlet to process the form data and display it back on the browser.
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AIM:

To create a simple web application that includes a form to accept a user's name and age, and then use a Servlet to process the submitted data. The Servlet will retrieve the form data, process it, and display the values back on the browser.

ALGORITHM:

1. Input:

- The user fills out a web form with their **name** and **age** and submits it.

2. Form Handling and Servlet Processing:

1. Create the Web Form:

- Create an HTML form with **two input fields**: one for the user's **name** and one for the user's **age**.
- Add a **submit button** to the form.
- Set the form's action attribute to the URL of the **Servlet** and the method to POST (or GET).

2. Create the Servlet Class:

- Write a Servlet that processes the **form data** submitted by the user.
- The Servlet will override the doPost() (or doGet()) depending on the form method) to handle form submissions.
- The Servlet will **retrieve the name and age** parameters from the request object using request.getParameter("name") and request.getParameter("age").

3. Process the Data in the Servlet:

- Extract the name and age values submitted from the form.
- **Validate the input** (e.g., check if the age is a valid number).
- Prepare a **response** that will display the user's **name** and **age** back on the browser.

4. Send Response to the Browser:

- The Servlet will send a **response** to the browser containing a message like:

- "Hello, [name]! You are [age] years old."

- The response will be displayed to the user in the browser.

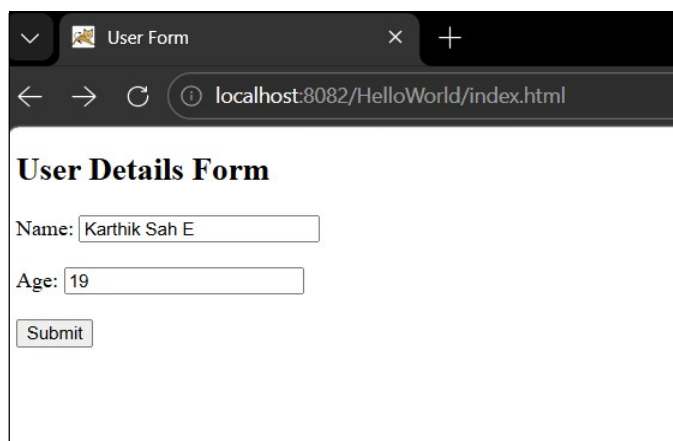
5. Deploy the Servlet:

- Compile the Servlet and deploy it to a **Servlet container** (like **Apache Tomcat**).
- Ensure the web.xml file or Servlet annotations are used to map the Servlet to the URL defined in the form's action attribute.

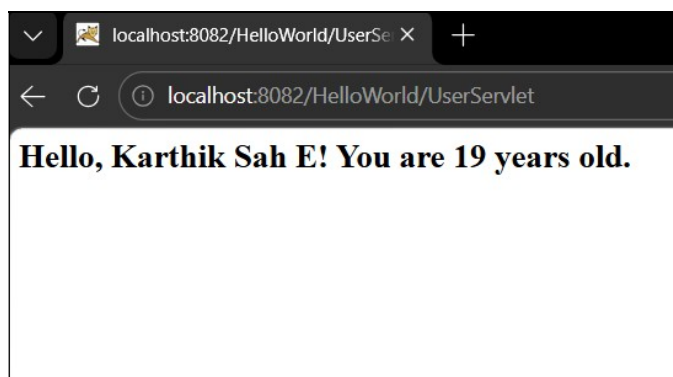
6. Test the Application:

- Open the web browser, fill in the form with the name and age, and submit it.
- The Servlet will process the data and display the message back to the user in the browser.

Output:



A screenshot of a web browser window. The title bar shows 'User Form'. The address bar shows 'localhost:8082/HelloWorld/index.html'. The page content is titled 'User Details Form'. It contains two input fields: 'Name: Karthik Sah E' and 'Age: 19'. Below these fields is a 'Submit' button.



A screenshot of a web browser window. The title bar shows 'localhost:8082/HelloWorld/UserServlet'. The address bar shows 'localhost:8082/HelloWorld/UserServlet'. The page content displays the message: 'Hello, Karthik Sah E! You are 19 years old.'

HTML File (index.html):

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>

    <title>User Form</title>

</head>

<body>

    <h2>User Details Form</h2>

    <form action="UserServlet" method="post">

        Name: <input type="text" name="name" required><br><br>

        Age: <input type="text" name="age" required><br><br>

        <input type="submit" value="Submit">

    </form>

</body>

</html>
```

Java File (UserServlet.java) :

```
import java.io.*;

import jakarta.servlet.*;

import jakarta.servlet.http.*;

import jakarta.servlet.annotation.WebServlet;

public class UserServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request,

                           HttpServletResponse response)

        throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();
```

```

String name = request.getParameter("name");

String ageStr = request.getParameter("age");


try {

    int age = Integer.parseInt(ageStr);


    out.println("<html><body>");

    out.println("<h2>Hello, " + name +

        "! You are " + age + " years old.</h2>");

    out.println("</body></html>");


} catch (NumberFormatException e) {

    out.println("<html><body>");

    out.println("<h2>Invalid age entered.</h2>");

    out.println("</body></html>");

}

}

}

```

Java Compiled File (UserServlet.class) :

// Source code is decompiled from a .class file using FernFlower decompiler (from IntelliJ IDEA).

```

import jakarta.servlet.ServletException;

import jakarta.servlet.http.HttpServlet;

import jakarta.servlet.http.HttpServletRequest;

import jakarta.servlet.http.HttpServletResponse;

import java.io.IOException;

import java.io.PrintWriter;

```

```

public class UserServlet extends HttpServlet {

    public UserServlet() {

    }

    protected void doPost(HttpServletRequest var1, HttpServletResponse var2) throws
    ServletException, IOException {

        var2.setContentType("text/html");

        PrintWriter var3 = var2.getWriter();

        String var4 = var1.getParameter("name");

        String var5 = var1.getParameter("age");

        try {

            int var6 = Integer.parseInt(var5);

            var3.println("<html><body>");

            var3.println("<h2>Hello, " + var4 + "! You are " + var6 + " years old.</h2>");

            var3.println("</body></html>");

        } catch (NumberFormatException var7) {

            var3.println("<html><body>");

            var3.println("<h2>Invalid age entered.</h2>");

            var3.println("</body></html>");

        }

    }

}

```

File (web.xml) :

```

<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"

    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

```

```
xsi:schemaLocation="https://jakarta.ee/xml/ns/jakartaee
https://jakarta.ee/xml/ns/jakartaee/web-app_6_0.xsd"
version="6.0">
```

```
<servlet>
  <servlet-name>HelloWorld</servlet-name>
  <servlet-class>HelloWorldServlet</servlet-class>
</servlet>
```

```
<servlet-mapping>
  <servlet-name>HelloWorld</servlet-name>
  <url-pattern>/hello</url-pattern>
</servlet-mapping>
```

```
<servlet>
  <servlet-name>UserServlet</servlet-name>
  <servlet-class>UserServlet</servlet-class>
</servlet>
```

```
<servlet-mapping>
  <servlet-name>UserServlet</servlet-name>
  <url-pattern>/UserServlet</url-pattern>
</servlet-mapping>
```

```
</web-app>
```

RESULT:

Thus A Servlet was created to process the HTML form data and display it back on the browser.